

Contributing to Rev80

Dev environment setup, project layout, and build/packaging instructions. For usage, installation, configuration, and the vibration/DSP technical reference, see [README.md](#).

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Development environment

Linux / macOS recommended.

```
git clone <repo-url>/vibegui.git
cd vibegui

# Create and activate a virtual environment
python -m venv .venv
source .venv/bin/activate          # Windows: .venv\Scripts\activate

# Install in editable mode with dev tools
pip install -e ".[dev]"

# Enable the repo's git hooks (once per clone) – keeps doc/*.pdf in sync
# with README.md, CONTRIBUTING.md, doc/PROGRESS.md, doc/CHANGELOG.md on commit
git config core.hooksPath .githubhooks

# Run the test suite (no hardware required – uses SimulatedSensor)
pytest tests/

# Launch the app against the source tree
python -m rev80
```

Rendering docs to PDF

`doc/README.pdf` , `doc/CONTRIBUTING.pdf` , `doc/PROGRESS.pdf` , and `doc/CHANGELOG.pdf` are committed alongside their Markdown sources. The same `pre-commit` hook that runs `ruff` (`.githubhooks/pre-commit` , enabled by the `git config core.hooksPath .githubhooks` step above) also re-renders and stages the PDF for any of those four files that are part of a commit, via `scripts/render_md.sh` :

```
# Render one file manually, e.g. after editing without committing
./scripts/render_md.sh README.md          # → doc/README.pdf
./scripts/render_md.sh doc/CHANGELOG.md    # → doc/CHANGELOG.pdf
```

Requires `pandoc` and `weasyprint` on PATH. If either is missing, the hook prints a warning and skips rendering rather than blocking the commit — install them to keep the PDFs current:

```
# Debian/Ubuntu
sudo apt-get install pandoc
pip install weasyprint
```

Project layout

src/rev80/	Python package
_paths.py	Runtime-safe path resolution (dev vs frozen)
_pico_loader.py	Windows DLL search path setup for frozen builds
logging.yaml	Logging configuration (bundled with package)
monitor/	Monitor Mode package
__init__.py	
session.py	MonitorSession dataclass
gate.py	IntervalGate scheduler
anomaly.py	AnomalyHook protocol; Rms/Spectral/FixedThreshold hooks;
Composite	
writer.py	MonitorWriterThread (daemon, writes session.h5)
controller.py	MonitorController (is_recording; hook supplied at start)
assets/	Package data – bundled into wheels and frozen builds alike
fonts/	CommitMono Nerd Font (gitignored; fetched by scripts/
fetch_font.sh)	
icons/hicolor/	App icon PNG set (16..256px), for `rev80 --install-desktop-
entry`	
	on Linux – see "Replacing the app icon" below
desktop.py	Linux .desktop launcher entry + icon install/uninstall
assets/	App icon source (edited directly by scripts/make_icons.sh)
rev80.svg	Source artwork
rev80.ico	Multi-resolution icon used by installer and exe
drivers/	PicoScope DLLs (Windows build only, not committed)
installer/	Inno Setup script
tests/	pytest suite
test_monitor_controller.py	
test_monitor_gate.py	
test_monitor_index.py	
test_monitor_session_load.py	full write→load→browse integration tests
build/	
rev80.spec	PyInstaller build spec
collect_pico_dlls.py	Collects PicoScope DLLs into drivers/
scripts/	
build.sh	Full build pipeline (run from Git Bash)
make_icons.sh	Regenerates rev80.ico + the hicolor PNG set via Inkscape +
ImageMagick	
fetch_font.sh	Downloads the CommitMono Nerd Font into src/rev80/assets/fonts/
render_md.sh	Renders a Markdown file to PDF (pandoc + weasyprint) – see doc/
.githooks/	
pre-commit	ruff check (blocking; scope matches CI), and re-renders doc/
*.pdf	for any of README.md, CONTRIBUTING.md, doc/PROGRESS.md,

```
doc/CHANGELOG.md staged in the commit
doc/
  README.pdf, CONTRIBUTING.pdf, PROGRESS.pdf, CHANGELOG.pdf    Rendered by the pre-commit
hook
  PROGRESS.md          Delivery history – client requirements mapped to commits
  CHANGELOG.md         Keep-a-Changelog-format change log
```

Testing

Tests use `VibeSensor.simulated()` and synthetic data — no hardware required. The full suite runs in ~50 s. Tests that write `.h5` scratch files use a project-relative `./DEVDATA/` directory (hardcoded in the test files, independent of the app's own `~/Documents/Rev80/data/` save location) — safe to delete between runs.

```
# Run all tests (no hardware)
pytest tests/

# Skip hardware-dependent tests explicitly
pytest tests/ -k "not hardware and not siggen"

# Single file
pytest tests/test_monitor_session_load.py

# Single test
pytest tests/test_vibechecker.py::test_save
```

Test file	Coverage
test_sample.py	AcquisitionSettings , VibeSample , ChannelResult
test_vibechecker.py	DataCollector stream lifecycle, save/load, trend
test_acquisition_settings.py	Derived properties, setter validation
test_scope_sensor.py	Sensor calibration pipeline, mV → EU scaling
test_picoscope.py	FindPicoScope enumeration logic (mocked driver)
test_config.py	YAML persistence, defaults, atomic writes
test_monitor_gate.py	IntervalGate snap-to-grid, burst entry/exit
test_monitor_anomaly.py	RmsThresholdHook (warmup, streak, t=0 tracking), SpectralThresholdHook (EWMA, band masking), FixedThresholdHook (unit conversion, mV skip, modality mismatch), CompositeAnomalyHook
test_monitor_controller.py	MonitorController lifecycle, hook wiring, cooldown gating, HDF5 structure
test_monitor_session_load.py	Full write → load → browse integration: interval frames, burst frames, trend reconstruction, NaN regression guard

Hardware-specific tests skip automatically when no PicoScope is detected.

Replacing the app icon

Edit `assets/icons/rev80.svg`, then regenerate both the Windows `.ico` and the Linux `hicolor` PNG set from it — no changes to `build/rev80.spec`, `installer/rev80.iss`, or `desktop.py` are needed either way:

```
# From the repo root — requires inkscape and imagemagick
./scripts/make_icons.sh
```

This updates `src/rev80/assets/icons/hicolor/*/apps/rev80.png` (Linux desktop entry) in place — commit it. It currently writes `rev80.ico` to the repo root rather than `assets/icons/rev80.ico`; move it there manually before committing so `build/rev80.spec` picks up the update.

Building the Windows Installer

Produces a self-contained one-directory executable and a standalone installer (`.exe`) via PyInstaller and Inno Setup. **The build must run on a 64-bit Windows machine.**

Prerequisites

Tool	Where to get it	Notes
Python 3.10+ (64-bit)	python.org	Must be 64-bit; add to PATH
Git for Windows	git-scm.com	Provides Git Bash for <code>scripts/build.sh</code>
PicoSDK 11.1.418	picotech.com/downloads	Reboot after install
Inno Setup 6	jrsoftware.org/isinfo.php	Per-user install to <code>%LOCALAPPDATA%</code> is fine

Install project dependencies (from Git Bash or `cmd.exe`):

```
pip install -e ".[dev]"
```

Running the build

```
# Run from the repo root
# Full pipeline: collect DLLs → PyInstaller → Inno Setup
./scripts/build.sh

# Individual steps
./scripts/build.sh dlls           # collect PicoScope DLLs into drivers/ only
./scripts/build.sh pyinstaller    # PyInstaller only (skips DLL collection)
./scripts/build.sh installer      # Inno Setup only (requires dist/ to exist)
```

Output

Path	Description
<code>dist/rev80/rev80.exe</code>	Standalone executable (no install needed)
<code>installer/Output/Rev80Setup-<version>.exe</code>	Installer with Start Menu shortcut and uninstaller

Known constraints

- **64-bit only** — PicoSDK DLLs are 64-bit; 32-bit Python will not work.
- **DearPyGui pinned to 2.0.0** — versions above 2.0.0 have a known viewport crash on Windows.
- **USB kernel driver** — `ps4000a.dll` is the user-mode library; the USB kernel driver is installed separately by PicoSDK. Reboot required before first hardware connection.
- **Code signing** — the installer and executable are unsigned; Windows SmartScreen will warn on first run. Sign with `osslsigncode` and a certificate if you need this.
- **No CI/dedicated build server** — this build only runs on real Windows hardware (PyInstaller cross-compilation isn't reliable, and PicoSDK's DLLs are Windows-only), so it's currently a manual step: boot into Windows, pull, run `scripts/build.sh`. There's no GitHub Actions workflow for it.